

Freya Ashcombe

Research Scientist | Machine Learning & Supply Chain Optimization

Portland, OR 97204 | (503) 847-2156 | f.ashcombe@academic-email.com

EDUCATION

PhD in Computer Science, University of Oregon, Eugene, OR — 2024

- Dissertation: "Neural Architectures for Demand Forecasting in Multi-Echelon Supply Networks"
- GPA: 3.91/4.0

BSc in Mathematics with Computer Science, University of Manchester, UK — 2020

PROFESSIONAL EXPERIENCE

Research Scientist, Institute for Logistics Technology, Portland, OR — July 2024–Present

- Developed graph neural network architecture for supply chain disruption prediction; evaluated on benchmark dataset of 15,000+ real-world inventory trajectories achieving 23% MAPE improvement over baseline LSTM
- Implemented reinforcement learning framework for warehouse routing optimization; published results in *Journal of Operations Research Quarterly* (2025, 12 citations to date)
- Created evaluation pipeline for multi-objective optimization algorithms across three public supply chain datasets; released benchmark code and results via GitHub with 340+ stars
- Collaborated with domain experts on problem formulation; conducted three workshops with supply chain researchers to validate model assumptions and dataset selection
- Prototyped transformer-based model for lead time prediction in manufacturing networks; manuscript under review at *International Conference on Machine Learning*

PhD Candidate, University of Oregon, Eugene, OR — September 2020–June 2024

- Investigated attention mechanisms for time-series forecasting in logistics networks; published "Temporal Attention in Supply Chain Demand Prediction" in *Proceedings of NeurIPS 2023* (47 citations, venue rank: top-tier)
- Conducted extensive ablation studies on sequence length, attention head configurations, and feature engineering approaches using proprietary academic dataset of 5,000+ supply chain transactions
- Developed simulation environment modeling multi-node distribution systems; enabled reproducible benchmarking of forecasting algorithms
- Presented research at 6 international conferences including ICML, NeurIPS, and INFORMS Annual Meeting
- Performed literature review synthesizing 200+ papers on demand forecasting, inventory optimization, and neural network architectures for sequential data
- Recipient of Graduate Research Fellowship in Machine Learning for Sustainability (2021–2023)

Research Intern, Distributed Systems Lab, University of Manchester, UK — May 2019–August 2020

- Analyzed machine learning approaches for time-series anomaly detection in manufacturing environments
- Evaluated three deep learning architectures (LSTM, GRU, Temporal CNN) on benchmark datasets from UCI and Kaggle repositories
- Contributed to paper "Comparative Analysis of Neural Architectures for Industrial Time Series" presented at *IEEE International Conference on Data Mining* (2020, 8 citations)
- Assisted in designing data collection protocols and exploratory data analysis workflows

PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Publications:

- Ashcombe, F., Reeves, D., Kahn, S. (2025). "Graph Neural Networks for Supply Chain Disruption Detection." *Journal of Operations Research Quarterly*, 52(4), 891–912. [12 citations]
- Ashcombe, F., Patel, N. (2023). "Temporal Attention in Supply Chain Demand Prediction." *Proceedings of NeurIPS 2023*, New Orleans, LA. [47 citations]
- Chen, R., Ashcombe, F., Morley, P. (2020). "Comparative Analysis of Neural Architectures for Industrial Time Series." *Proceedings of IEEE ICDM 2020*, Shenyang, China. [8 citations]

Conference Presentations:

- NeurIPS 2023 (Poster), INFORMS Annual Meeting 2024 (Talk), ICML 2024 Workshop on Sustainability (Spotlight)

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, Julia
- **ML/AI Frameworks:** PyTorch, TensorFlow, JAX, scikit-learn
- **Architectures:** Transformers, Graph Neural Networks, LSTMs, Temporal Convolutional Networks, Reinforcement Learning (PPO, DQN)
- **Data & Tools:** Pandas, NumPy, Weights & Biases, Git, Jupyter, HuggingFace Transformers
- **Domains:** Time-series forecasting, demand planning, inventory optimization, anomaly detection, supply chain simulation

AWARDS & RECOGNITION

- Graduate Research Fellowship in Machine Learning for Sustainability — University of Oregon (2021–2023)
- Best Paper Runner-up — INFORMS Sustainability & Analytics Track (2024)
- Dean's List — University of Manchester (2018–2020)